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APPLICATION LAYER

Contents:Client Server Paradigm: Communication using TCP and UDP, Peer to Peer Paradigm, Application Layer

Protocols: DNS, FTP, TFTP, HTTP, SMTP, POP, IMAP, MIME, Network

Management: SNMP.

***Unit Objectives and outcomes:** On completion the students will be able to:*

To introduce application layer services.

Unit outcomes : On completion the students will be able to

Understand services and protocols used at application layer.

Outcome Mapping:

PEOs:

POs:

COs: CO3

PSOs:

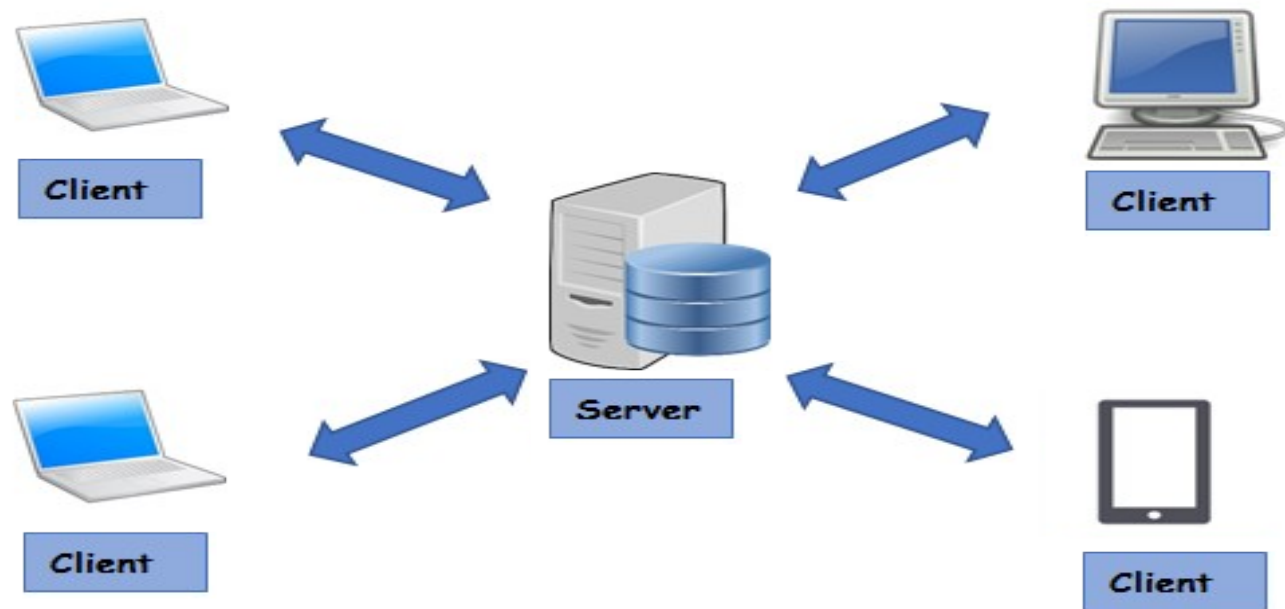
Introduction to application layer

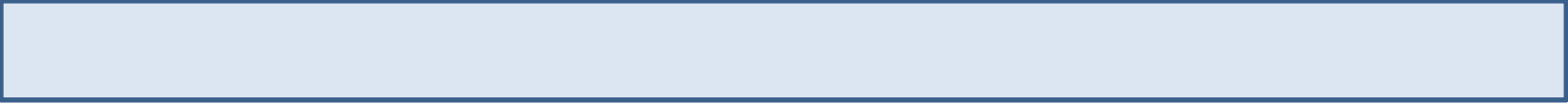
- Application layer is the top most layer in OSI and TCP/IP layered model. This layer exists in both layered Models because of its significance, of interacting with user and user applications.
- This layer is for applications which are involved in communication system.
- A user may or may not directly interacts with the applications. Application layer is where the actual communication is initiated and reflects.
- Because this layer is on the top of the layer stack, it does not serve any other layers.
- Application layer takes the help of Transport and all layers below it to communicate or transfer its data to the remote host.

Functions of Application Layer

- **Mail Services:** This layer provides the basis for E-mail forwarding and storage.
- **Network Virtual Terminal:** It allows a user to log on to a remote host. The application creates software emulation of a terminal at the remote host. User's computer talks to the software terminal which in turn talks to the host and vice versa. Then the remote host believes it is communicating with one of its own terminals and allows user to log on.
- **Directory Services:** This layer provides access for global information about various services.
- **File Transfer, Access and Management (FTAM):** It is a standard mechanism to access files and manages it. Users can access files in a remote computer and manage it. They can also retrieve files from a remote computer.

Client-server architecture: An application program running on the local machine sends a request to another application program is known as a client, and a program that serves a request is known as a server. For example, when a web server receives a request from the client host, it responds to the request to the client host.





➤The client-server model, or client-server architecture, is a distributed application framework dividing tasks between servers and clients, which either reside in the same system or communicate through a computer network or the Internet.

➤The client relies on sending a request to another program in order to access a service made available by a server.

➤The server runs one or more programs that share resources with and distribute work among clients.

➤The client server relationship communicates in a request–response messaging pattern and must adhere to a common communications protocol, which formally defines the rules, language, and dialog patterns to be used.

➤Client-server communication typically adheres to the TCP/IP protocol suite.

Peer to Peer Architecture

Peer to Peer Architecture In the common client-server architecture, multiple clients will communicate with a central server. A peer-to-peer (P2P) architecture consists of a decentralized network of peers - nodes that are both clients and servers.

P2P networks distribute the workload between peers, and all peers contribute and consume resources within the network without the need for a centralized server.

However, not all peers are necessarily equal. Super peers may have more resources and can contribute more than they consume.

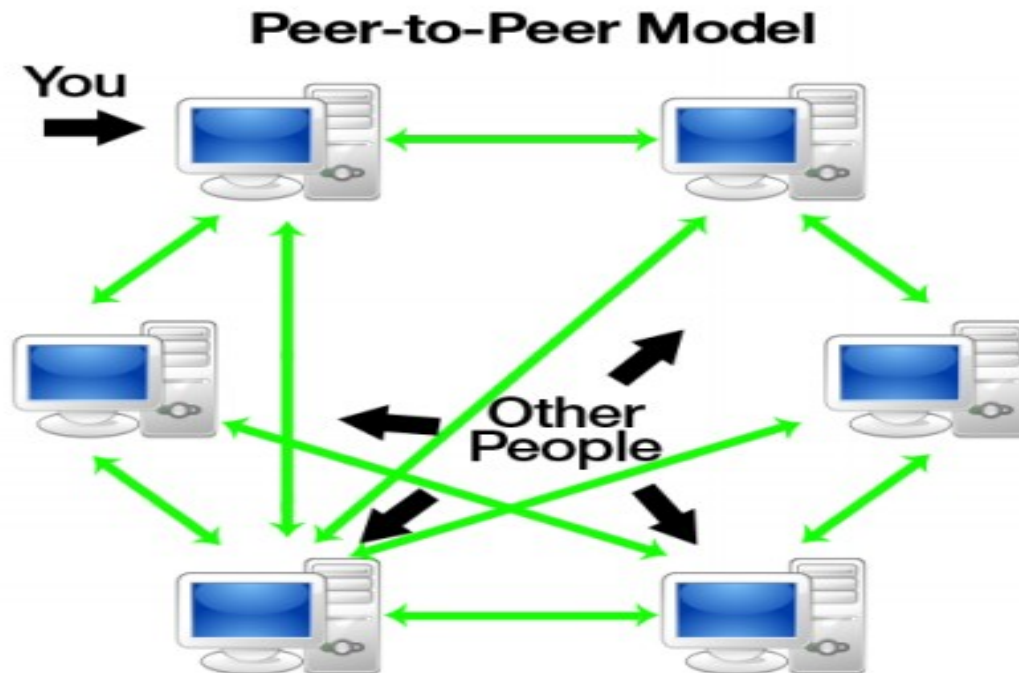
Edge peers do not contribute any resources, they only consume from the network.

In its purest form, P2P architecture is completely decentralized. However, in application, sometimes there is a central tracking server layered on top of the P2P network to help peers find each other and manage the network.

Features of P2P architecture

Self scalability: In a file sharing system, although each peer generates a workload by requesting the files, each peer also adds a service capacity by distributing the files to the peer.

Cost-effective: It is cost-effective as it does not require significant server infrastructure and server bandwidth.



Client-Server Network

In Client-Server Network, Clients and server are differentiated, Specific server and clients are present.

Client-Server Network focuses on information sharing.

In Client-Server Network, Centralized server is used to store the data.

In Client-Server Network, Server respond the services which is request by Client.

Client-Server Network are costlier than Peer-to-Peer Network.

Client-Server Network are more stable than Peer-to-Peer Network.

Client-Server Network is used for both small and large networks.

Peer-to-Peer Network

In Peer-to-Peer Network, Clients and server are not differentiated.

While Peer-to-Peer Network focuses on connectivity.

While in Peer-to-Peer Network, Each peer has its own data.

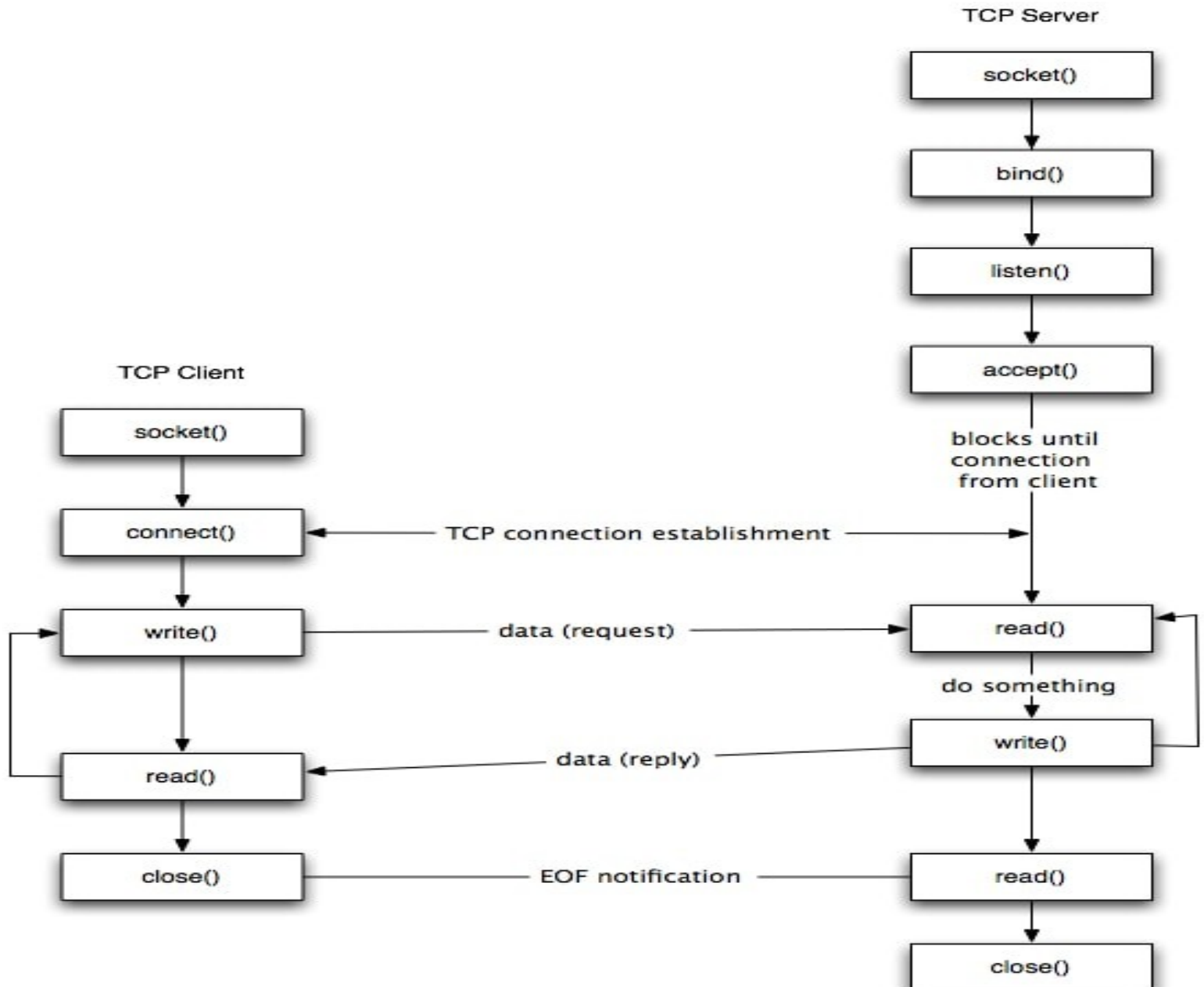
While in Peer-to-Peer Network, Each and every node can do both request and respond for the services.

While Peer-to-Peer Network are less costlier than Client-Server Network.

While Peer-to-Peer Network are less stable if number of peer is increase.

While Peer-to-Peer Network is generally suited for small networks with fewer than 10 computers.

communication using tcp



Steps to create a client using TCP/IP

Create a socket using the `socket()` **function** .

Initialize the socket address structure as per the server and connect the socket to the address of the server using the `connect()`;

Receive and send the data using the `recv()` and `send()` functions.

Close the connection by calling the `close()` function.

Steps to create a server using TCP/IP

Create a socket using the `socket()` function

Initialize the socket address structure and bind the socket to an address using the `bind()` function.

Listen for connections with the `listen()` function.

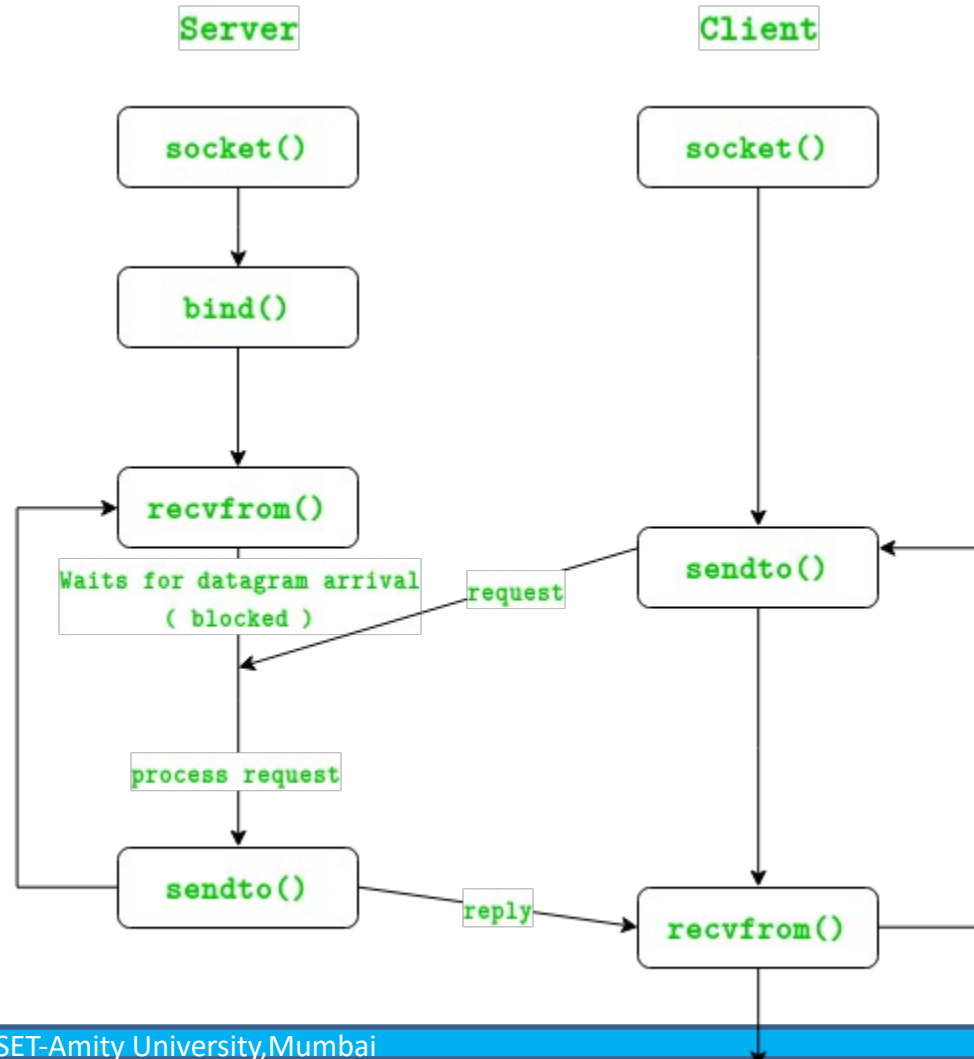
Accept a connection with the `accept()` function system call. This call typically blocks until a client connects to the server.

Receive and send data by using the `recv()` and `send()` function.

Close the connection by using the `close()` function.

UDP Server-Client implementation

In UDP, the client does not form a connection with the server like in TCP and instead just sends a datagram. Similarly, the server need not accept a connection and just waits for datagrams to arrive. Datagrams upon arrival contain the address of sender which the server uses to send data to the correct client.



The entire process can be broken down into following steps :

UDP Server :

- 1.Create UDP socket.
- 2.Bind the socket to server address.
- 3.Wait until datagram packet arrives from client.
- 4.Process the datagram packet and send a reply to client.
- 5.Go back to Step 3.

UDP Client :

- 1.Create UDP socket.
- 2.Send message to server.
- 3.Wait until response from server is recieved.
- 4.Process reply and go back to step 2, if necessary.
- 5.Close socket descriptor and exit.

Application Layer Protocols

At the top of the TCP/IP protocol architecture is the Application Layer. This layer includes all processes that use the Transport Layer protocols to deliver data.

There are many applications protocols. Most provide user services, and new services are always being added to this layer.

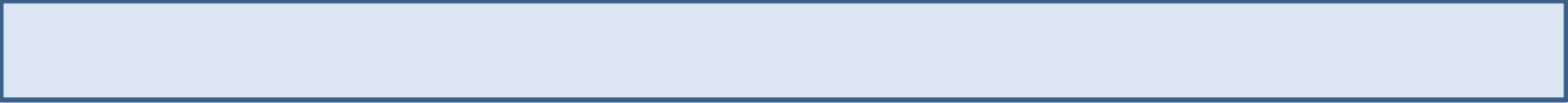
The most widely known and implemented applications protocols are:

- Telnet, the Network Terminal Protocol, provides remote login over the network
- FTP, the File Transfer Protocol, is used for interactive file transfer
- SMTP, the Simple Mail Transfer Protocol delivers electronic mail
- DNS, An application layer protocol defines how the application processes running on different systems, pass the messages to each other.

DNS (Domain Name System)

- An application layer protocol defines how the application processes running on different systems, pass the messages to each other.
- DNS is a directory service that provides a mapping between the name of a host on the network and its numerical address.
- DNS is required for the functioning of the internet.
- Each node in a tree has a domain name, and a full domain name is a sequence of symbols specified by dots.
- DNS is a service that translates the domain name into IP addresses.
- This allows the users of networks to utilize user-friendly names when looking for other hosts instead of remembering the IP addresses.

For example, suppose the FTP site at EduSoft had an IP address of 132.147.165.50, most people would reach this site by specifying ftp.EduSoft.com. Therefore, the domain name is more reliable than IP address.



Any DNS following three major components-

Domain Name Space and Resource Records:

These are specifications for a tree-structured name space along with data associated with these names.

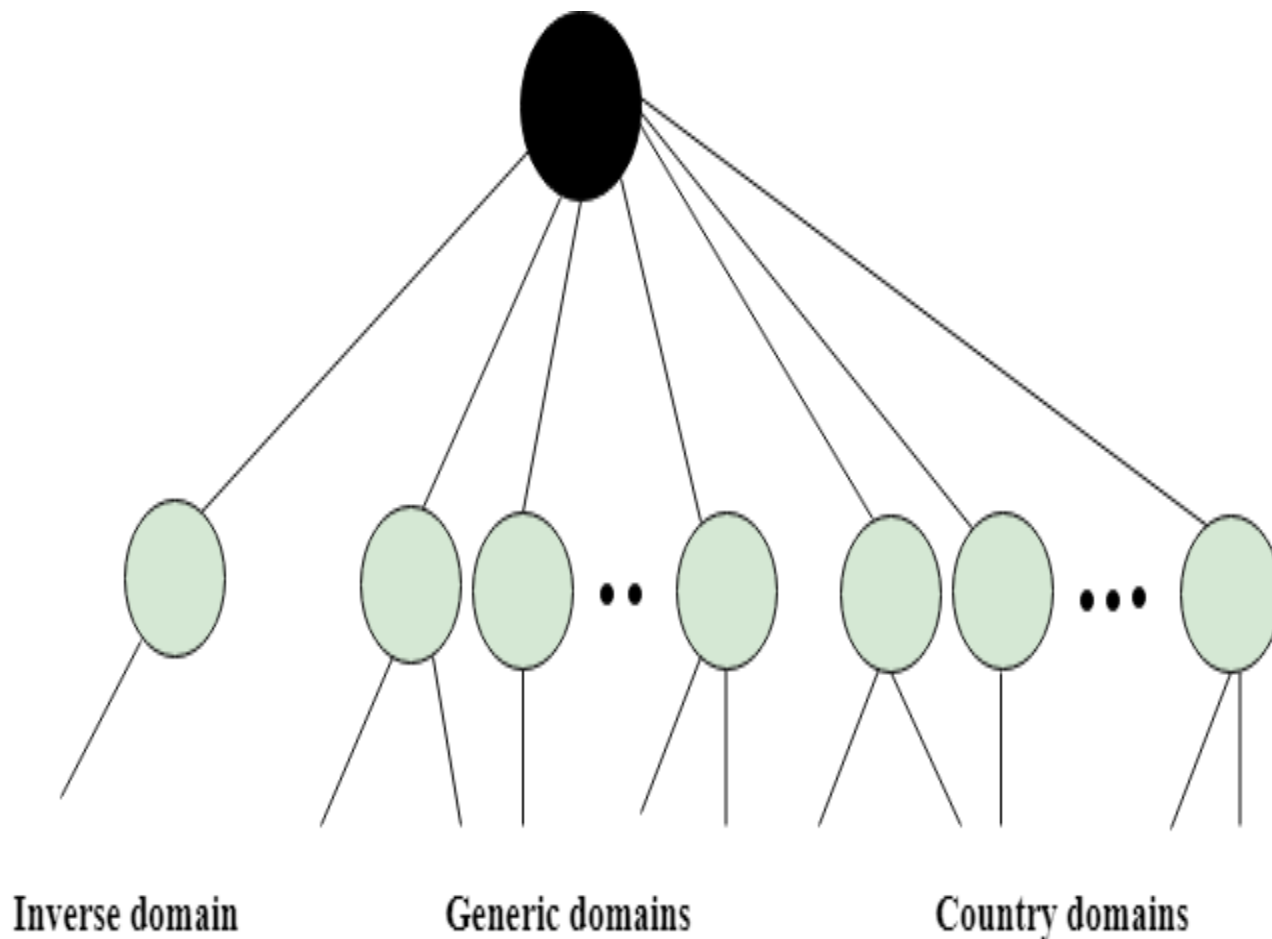
Name Servers:

These are server programs that contain information regarding the domain tree's structure along with set information.

Resolvers:

These are programs that are used for extracting information from name servers in reply to the requests done by the clients. The resolvers need to access at least one name server and use the information of that name server for answering a query directly.

Domain Name Space :- DNS is a TCP/IP protocol used on different platforms. The domain name space is divided into three different sections: generic domains, country domains, and inverse domain.



Generic Domains

It defines the registered hosts according to their generic behavior.

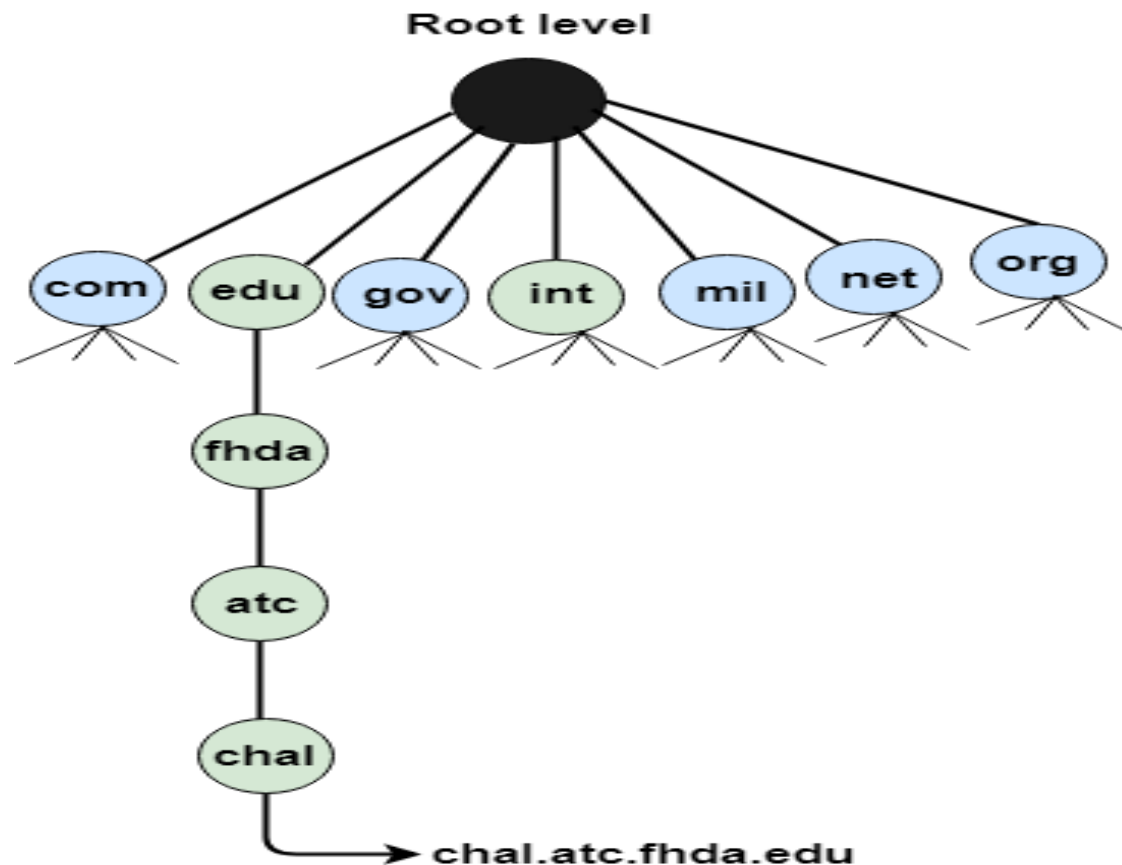
Each node in a tree defines the domain name, which is an index to the DNS database.

It uses three-character labels, and these labels describe the organization type.

Label	Description
aero	Airlines and aerospace companies
biz	Businesses or firms
com	Commercial Organizations
coop	Cooperative business Organizations
edu	Educational institutions
gov	Government institutions

Country Domain

The format of country domain is same as a generic domain, but it uses two-character country abbreviations (e.g., us for the United States) in place of three character organizational abbreviations.



Inverse Domain

The inverse domain is used for mapping an address to a name.

When the server has received a request from the client, and the server contains the files of only authorized clients.

To determine whether the client is on the authorized list or not, it sends a query to the DNS server and ask for mapping an address to the name.

DNS Resolver

A DNS resolver, also called a **recursive resolver**, is a server designed to receive DNS queries from web browsers and other applications. The resolver receives a hostname - for example, **www.example.com** - and is responsible for tracking down the IP address for that hostname.

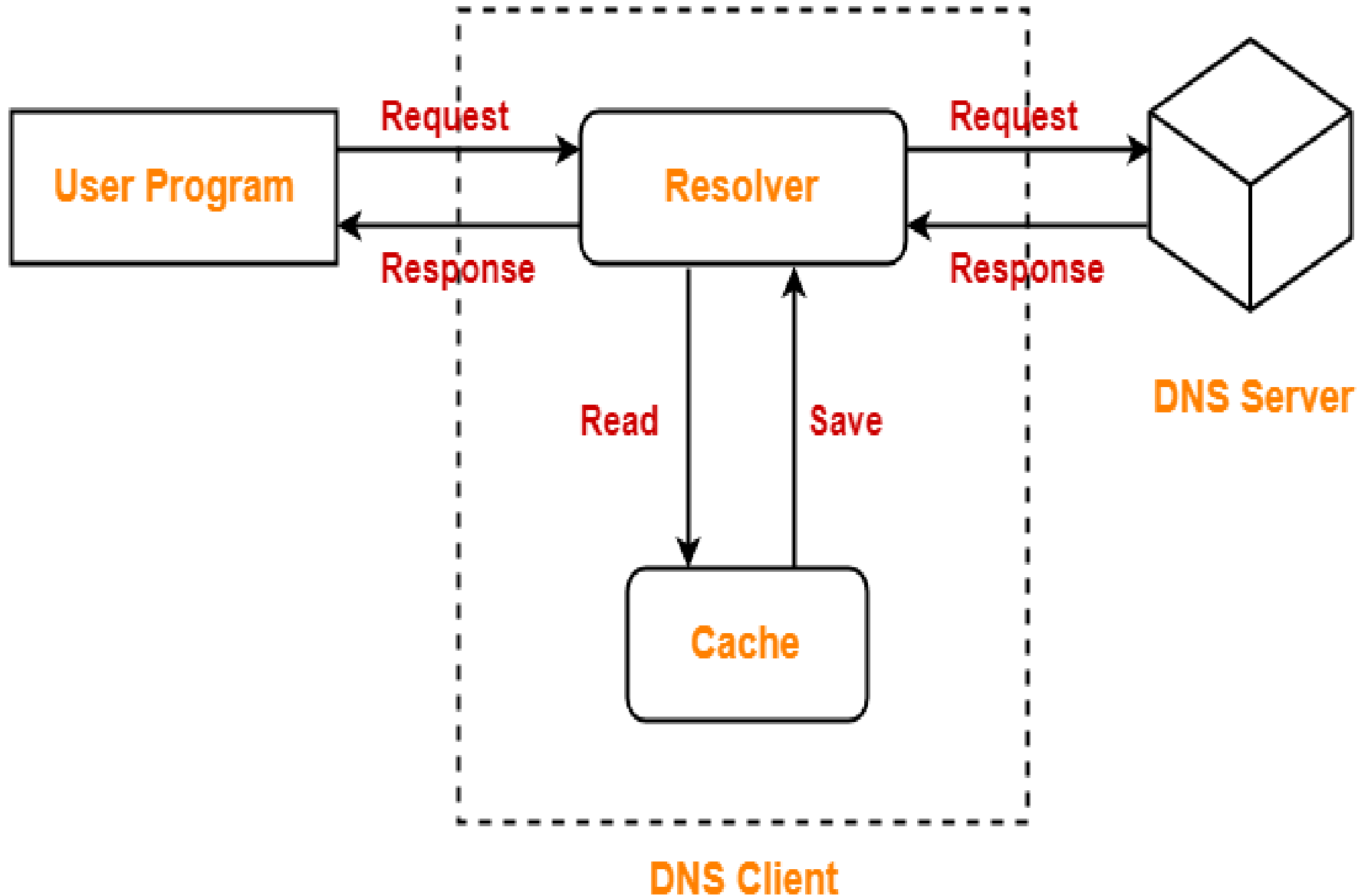
The DNS resolver might be operated by the local network, an Internet Service Provider (ISP), a mobile carrier, a WIFI network, or other third party.

The resolver starts by looking in its local cache or that of the operating system on the local device - if the hostname is found, it is resolved immediately.

If not found, the resolver contacts a **DNS Root Server** and receives details of a **TLD Name Server**.

Via the TLD Name Server, it receives details of an **Authoritative Name Server**, and asks it for the IP that matches the requested hostname. When it receives the IP, the query is resolved.

The following diagram illustrates the process of DNS resolution-



DNS Message Format

There is one DNS message defined for both queries and responses.

DNS has two types of messages: query and response.

Both types have the same format.

The query message consists of a header and question records; the response message consists of a header, question records, answer records, authoritative records, and additional records (see Figure4).

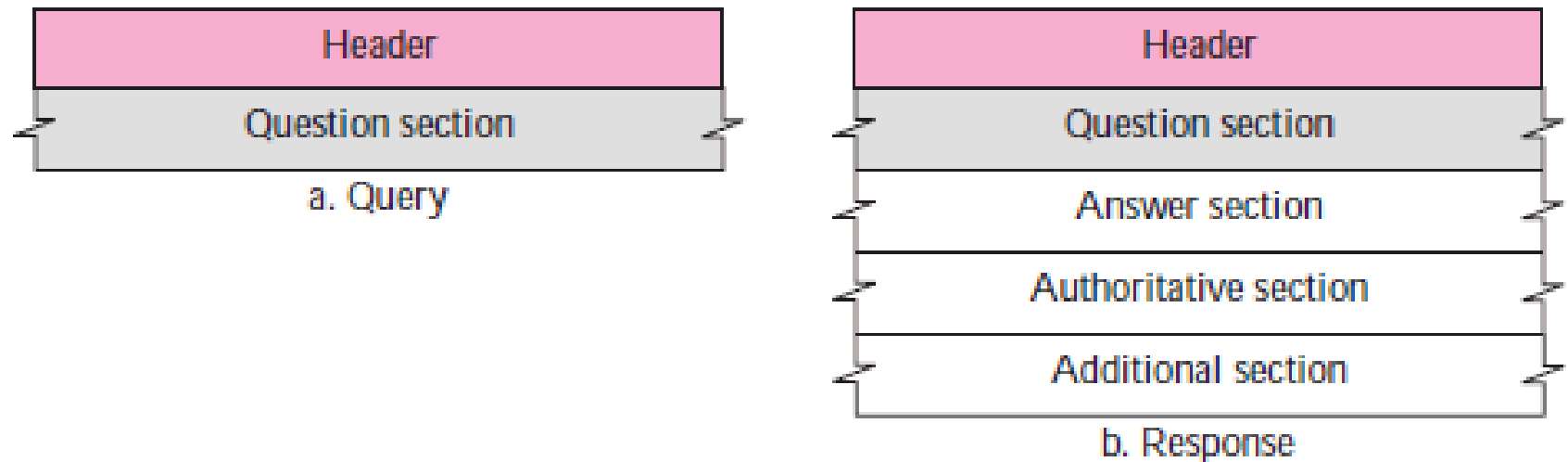


Fig4: Query and response messages

Header:

Both query and response messages have the same header format with some fields set to zero for the query messages. The header is 12 bytes and its format is shown in Figure5.

Identification	Flags
Number of question records	Number of answer records (All 0s in query message)
Number of authoritative records (All 0s in query message)	Number of additional records (All 0s in query message)

Identification: This is a 16-bit field used by the client to match the response with the query.

The client uses a different identification number each time it sends a query.

The server duplicates this number in the corresponding response.

Flags: This is a 16-bit field consisting of the subfields shown in Figure6.

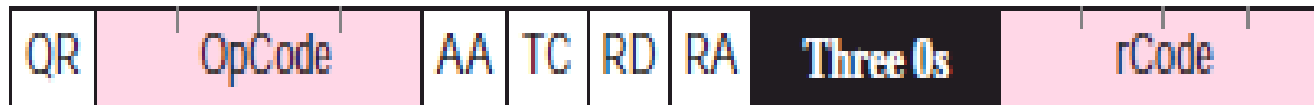


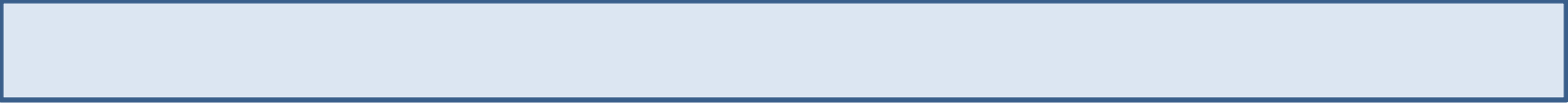
Fig6: Flags field

A brief description of each flag subfield follows.

QR (query/response): This is a 1-bit subfield that defines the type of message. If it is 0, the message is a query. If it is 1, the message is a response.

OpCode: This is a 4-bit subfield that defines the type of query or response (0 if standard, 1 if inverse, and 2 if a server status request).

AA (authoritative answer): This is a 1-bit subfield. When it is set (value of 1) it means that the name server is an authoritative server. It is used only in a response message.



TC (truncated): This is a 1-bit subfield. When it is set (value of 1), it means that the response was more than 512 bytes and truncated to 512. It is used when DNS uses the services of UDP (see Section 19.8 on Encapsulation).

RD (recursion desired): This is a 1-bit subfield. When it is set (value of 1) it means the client desires a recursive answer. It is set in the query message and repeated in the response message.

RA (recursion available): This is a 1-bit subfield. When it is set in the response, it means that a recursive response is available. It is set only in the response message.

Reserved: This is a 3-bit subfield set to 000.

rCode: This is a 4-bit field that shows the status of the error in the response.

Question Section:

This is a section consisting of one or more question records. It is present on both query and response messages.

Answer Section:

This is a section consisting of one or more resource records. It is present only on response messages. This section includes the answer from the server to the client (resolver).

Authoritative Section:

This is a section consisting of one or more resource records. It is present only on response messages. This section gives information (domain name) about one or more authoritative servers for the query.

Additional Information Section:

This is a section consisting of one or more resource records. It is present only on response messages. This section provides additional information that may help the resolver.

What is a DNS record?

DNS records (aka zone files) are instructions that live in authoritative DNS servers and provide information about a domain including what IP address is associated with that domain and how to handle requests for that domain.

These records consist of a series of text files written in what is known as DNS syntax. DNS syntax is just a string of characters used as commands that tell the DNS server what to do.

All DNS records also have a 'TTL', which stands for time-to-live, and indicates how often a DNS server will refresh that record.

A record - The record that holds the IP address of a domain.

CNAME record - Forwards one domain or subdomain to another domain, does NOT provide an IP address.

MX record - Directs mail to an email server.

TXT record - Lets an admin store text notes in the record.

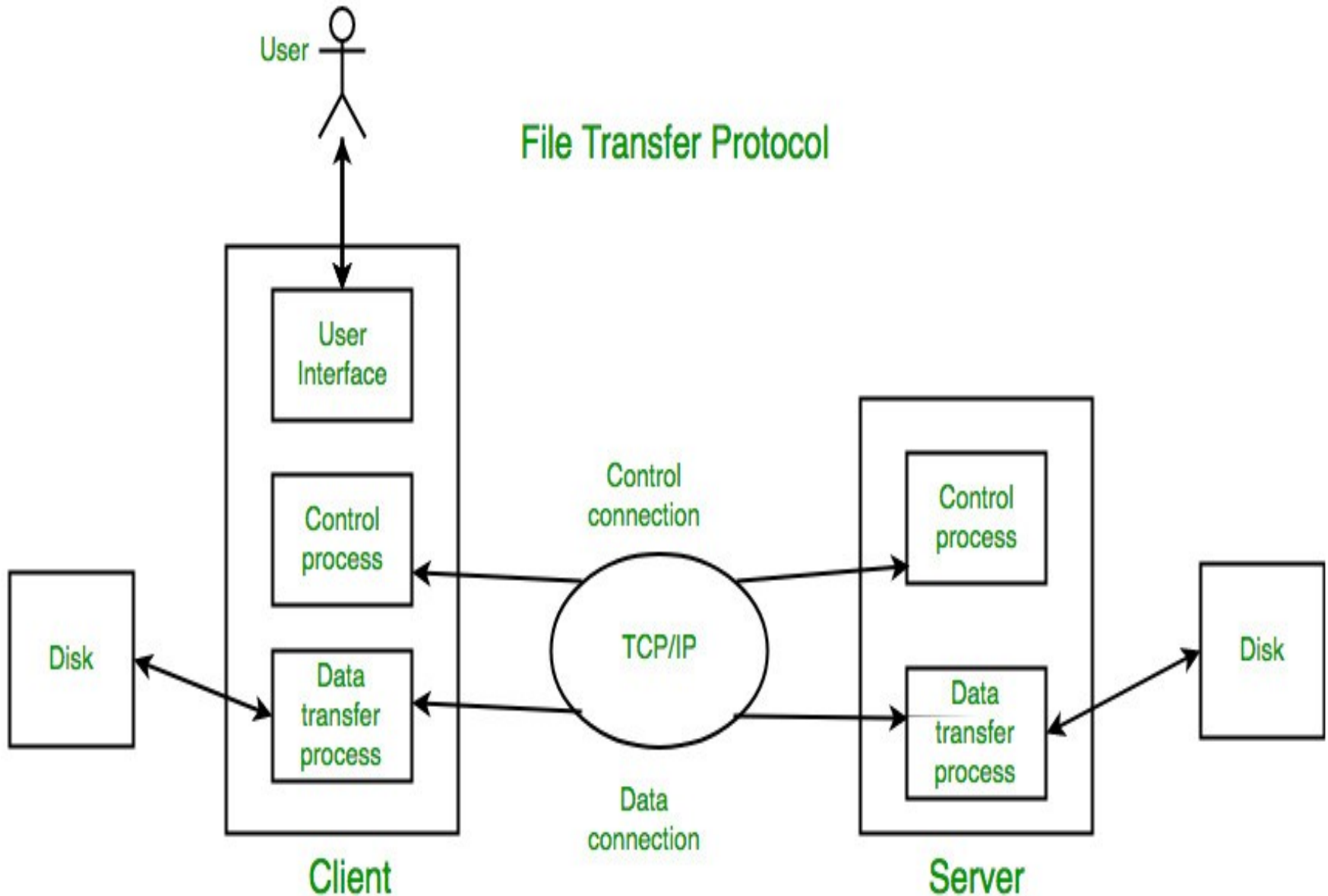
NS record - Stores the name server for a DNS entry.

SOA record - Stores admin information about a domain.

SRV record - Specifies a port for specific services.

PTR record - Provides a domain name in reverse-lookups.

File transfer protocol(FTP).



What is control connection?

For sending control information like user identification, password, commands to change the remote directory, commands to retrieve and store files, etc., FTP makes use of control connection. The control connection is initiated on **port number 21**.

What is data connection?

For sending the actual file, FTP makes use of data connection. A data connection is initiated on **port number 20**. Some protocols send their request and response header lines and the data in the same TCP connection. For this reason, they are said to send their control information in-band. HTTP and SMTP are such examples.

FTP Session :

When a FTP session is started between a client and a server, the client initiates a control TCP connection with the server side.

The client sends control information over this. When the server receives this, it initiates a data connection to the client side.

Only one file can be sent over one data connection. But the control connection remains active throughout the user session.

As we know HTTP is stateless i.e. it does not have to keep track of any user state. But FTP needs to maintain a state about its user throughout the session.

Data Structures : FTP allows three types of data structures :

File Structure – In file-structure there is no internal structure and the file is considered to be a continuous sequence of data bytes.

Record Structure – In record-structure the file is made up of sequential records.

Page Structure – In page-structure the file is made up of independent indexed pages.

FTP Commands –

USER – This command sends the user identification to the server.

PASS – This command sends the user password to the server.

CWD – This command allows the user to work with a different directory or dataset for file storage or retrieval without altering his login or accounting information.

RMD – This command causes the directory specified in the path-name to be removed as a directory.

MKD – This command causes the directory specified in the pathname to be created as a directory.

PWD – This command causes the name of the current working directory to be returned in the reply.

RETR – This command causes the remote host to initiate a data connection and to send the requested file over the data connection.

STOR – This command causes to store a file into the current directory of the remote host.

LIST – Sends a request to display the list of all the files present in the directory.

ABOR – This command tells the server to abort the previous FTP service command and any associated transfer of data.

QUIT – This command terminates a USER and if file transfer is not in progress, the server closes the control connection.

Transmission Modes of FTP

For transforming files across the internet connection, it uses three transmission modes.

1. Stream Mode

This is called a default mode. In this mode, data transforms from FTP to TCP in the form of stream bytes. Here TCP is responsible for fragmenting data into small segments.

If transforming data are already in the form of stream bytes, then the data connection is automatically closed. Otherwise, it is closed by the sender.

2. Block Mode

Block mode transforms data from one host to another in the form of blocks. Each block preceded by a 3-byte header. The first byte contains all the information about the block hence know for the description block. Other two blocks containing the size of the block in the form of bytes.

3. Compressed Mode

Compress mode is used in the case of big file size. If the file size is big, it cannot send over the internet connection because of the size limit. In Compressed mode, a large file is compressed into a small size and then sends over the internet.

FTP login:-

Ftp login uses normal username and password scheme for granting access.

The username is sent to the server using the USER command and the password is sent using Pass command.

This sequence is unencrypted “on the wire” so may be defenseless in case of sniffing attack.

If the information provided by the client is accepted by the server ,the server will send a greeting to the client and the session will commence.

If the server supports it users may log in without providing login credentials, but the same server may authorize only limited access for such sessions

Advantages of FTP clients

- Allows you to transfer multiple files as well as directories
- The ability to resume a transfer if the connection is lost
- The ability to add items to a “queue” to be uploaded/downloaded
- Many FTP clients have the ability to schedule transfers
- No size limitation on single transfers (browsers only allow up to 2 GB)
- Many clients have scripting capabilities through command line

FTP Disadvantages

- It's old, 45 years old roughly
- Difficult to use if you do not have an IT background
- Hard sell to employees with other file sharing alternatives
- Easy for inexperienced users to wipe out work with a single click
- Can't change ownership of a file
- Not a secure data transfer due to lack of encryption; must use secure alternative such as SFTP
- Passwords and usernames are in plain text making it simpler for an authorized user to gain access

Trivial File Transfer Protocol (TFTP)

- Trivial File Transfer Protocol (TFTP) is a simple protocol used for transferring files.
- TFTP uses the User Datagram Protocol (UDP) to transport data from one end to another. TFTP is mostly used to read and write files/mail to or from a remote server.
- File transfer is one of the most essential technologies for client/server and computer network infrastructures.
- Trivial File Transfer Protocol is very simple in design and has limited features as compared to File Transfer Protocol (FTP).
- TFTP provides no authentication and security while transferring files. As a result, it is usually used for transferring boot files or configuration files between machines in a local setup.
- Because of its simple design, it is rarely used interactively by users in a computer network.
- Its lack of security also makes it dangerous for use over the Internet.

TFTP messages come in five types:

- RRQ (read request) to request devices to read files.
- WRQ (write request) to request devices to write files.
- DATA to carry file blocks.
- ACK (acknowledge) to acknowledge received file blocks.
- ERROR to tell the sending device when an operation cannot be performed.

Each file block has an identification number so that ACK messages from target devices can correctly report back to sending devices if a block is received successfully.

The ERROR message can occur for various reasons, such as an invalid file name or a file that does not have read/write permissions for the target device.

How TFTP Works ?

- Trivial File Transfer Protocol uses UDP for communicating files, hence it establishes a connection generally by using port 69.
- Once the connection is established, the client generally requests RRQ or WRQ. Here, RRQ means Read Request, and WRQ stands for Write Request.
- A client generally requests for reading request if it wants only to read the file and generated a written request if he wants to write a particular file that exists on the server.
- Once this is done, then files are communicated in the form of small packets.
- These packets are 512 bytes. The file to be communicated is divided into small packets, where each packet consists of 512 bytes.
- Once a packet is communicated from server to client, the server waits to receive an acknowledgement from the client that the packet has been received. Once the acknowledgement is received, the server sends the next packet of 512 bytes.
- This is done till the last packet is communicated from server-side to client-side.

FTP

FTP stands for File Transfer Protocol.

The software of FTP is larger than TFTP.

FTP works on two ports: 20 and 21.

FTP services are provided by TCP.

The complexity of FTP is higher than TFTP.

There are many commands or messages in FTP.

FTP need authentication for communication.

FTP is generally suited for uploading and downloading of files by remote users.

TFTP

TFTP stands for Trivial File Transfer Protocol.

While software of TFTP is smaller than FTP.

While TFTP works on 69 Port number.

While TFTP services are provided by UDP.

While the complexity of TFTP is less than FTP complexity.

There are only 5 messages in TFTP.

While TFTP does not need authentication for communication.

While TFTP is mainly used for transmission of configurations to and from network devices.

What is HTTP?

The Hypertext Transfer Protocol (HTTP) is the foundation of the World Wide Web, and is used to load web pages using hypertext links.

HTTP is an application layer protocol designed to transfer information between networked devices and runs on top of other layers of the network protocol stack.

A typical flow over HTTP involves a client machine making a request to a server, which then sends a response message.

HTTP was invented alongside HTML to create the first interactive, text-based web browser: the original World Wide Web.

Today, the protocol remains one of the primary means of using the Internet.

Basic Features

There are three basic features that make HTTP a simple but powerful protocol:

HTTP is connectionless: The HTTP client, i.e., a browser initiates an HTTP request and after a request is made, the client waits for the response. The server processes the request and sends a response back after which client disconnect the connection.

HTTP is media independent: It means, any type of data can be sent by HTTP as long as both the client and the server know how to handle the data content.

HTTP is stateless: The server and client are aware of each other only during a current request. Afterwards, both of them forget about each other. Due to this nature of the protocol, neither the client nor the browser can retain information between different requests across the web pages.

Basic Architecture

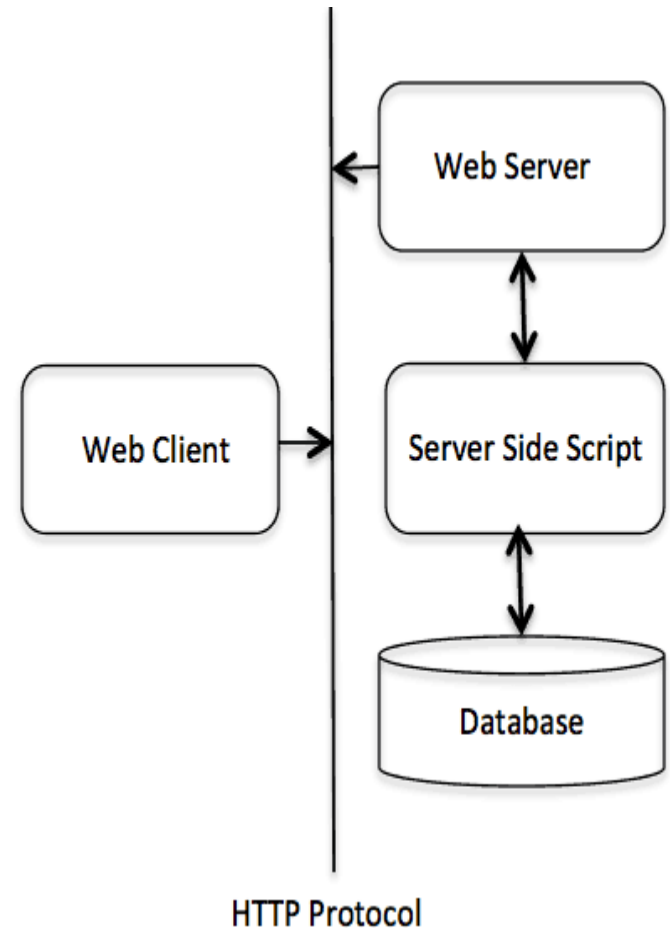
The following diagram shows a very basic architecture of a web application and depicts where HTTP sits: The HTTP protocol is a request/response protocol based on the client/server based architecture where web browsers, robots and search engines, etc. act like HTTP clients, and the Web server acts as a server.

Client

The HTTP client sends a request to the server in the form of a request method, URI, and protocol version, followed by a MIME-like message containing request modifiers, client information, and possible body content over a TCP/IP connection.

Server

The HTTP server responds with a status line, including the message's protocol version and a success or error code, followed by a MIME-like message containing server information, entity meta information, and possible entity-body content.



HTTP flow

When a client wants to communicate with a server, either the final server or an intermediate proxy, it performs the following steps:

- Open a TCP connection: The TCP connection is used to send a request, or several, and receive an answer. The client may open a new connection, reuse an existing connection, or open several TCP connections to the servers.
- Send an HTTP message: HTTP messages (before HTTP/2) are human-readable. With HTTP/2, these simple messages are encapsulated in frames, making them impossible to read directly, but the principle remains the same.
- Read the response sent by the server
- Close or reuse the connection for further requests.

Request Messages

A *request message* is sent by an agent to a server to request some action. These are the possible actions, called *methods*:

Method	Description
OPTIONS	A request for information about the options available.
GET	A request to retrieve information.
HEAD	Like a GET except that the server's response must not include an entity body; all of the header fields in the response are the same as if the entity body were present. This enables a client to get information about a resource without transferring the entity body.
POST	A request to accept the attached entity as a new subordinate to the identified URL.
PUT	A request to accept the attached entity and store it under the supplied URL. This may be a new resource with a new URL, or a replacement of the contents of an existing resource with an existing URL.
DELETE	Requests that the origin server delete a resource.
TRACE	Requests that the server return whatever is received as the entity body of the response. This can be used for testing and diagnostic purposes.

Response Messages

A *response message* is returned by a server to an agent in response to a request message. It may include an entity body containing hypertext-based information. In addition, the response message must specify a *status code*, which indicates the action taken on the corresponding request. Status codes are organized into the following categories:

Category	Description
Informational	The request has been received and processing continues. No entity body accompanies this response.
Successful	The request was successfully received, understood, and accepted.
Redirection	Further action is required to complete the request.
Client Error	The request contains a syntax error or the request cannot be fulfilled.
Server Error	The server failed to fulfill an apparently valid request.

BASIS FOR COMPARISON			HTTP	FTP
Basic	HTTP is used to access websites.		FTP transfers file from one one host to another.	
Connection	HTTP establishes data connection only.		FTP establishes two connection one for data and one for the control connection.	
TCP ports	HTTP uses TCP's port number 80.		FTP uses TCP's port number 20 and 21.	
URL	If you are using HTTP, http will appear in URL.		If you are using FTP, ftp will appear in URL.	
Efficient	HTTP is efficient in transferring smaller files like web pages.		FTP is efficient in transferring larger files.	
Authentication	HTTP does not require authentication.		FTP requires a password.	
Data	The content transferred to a device using HTTP is not saved to the memory of that device.		The file transferred to the host device using FTP is saved in the memory of that	

SMTP (Simple Mail Transfer Protocol.)

SMTP is a set of communication guidelines that allow software to transmit an electronic mail over the internet is called **Simple Mail Transfer Protocol**. It is a program used for sending messages to other computer users based on e-mail addresses.

It provides a mail exchange between users on the same or different computers, and it also supports:

- It can send a single message to one or more recipients.
- Sending message can include text, voice, video or graphics.
- It can also send the messages on networks outside the internet.

The main purpose of SMTP is used to set up communication rules between servers.

The servers have a way of identifying themselves and announcing what kind of communication they are trying to perform.

They also have a way of handling the errors such as incorrect email address.

Basic SMTP Architecture

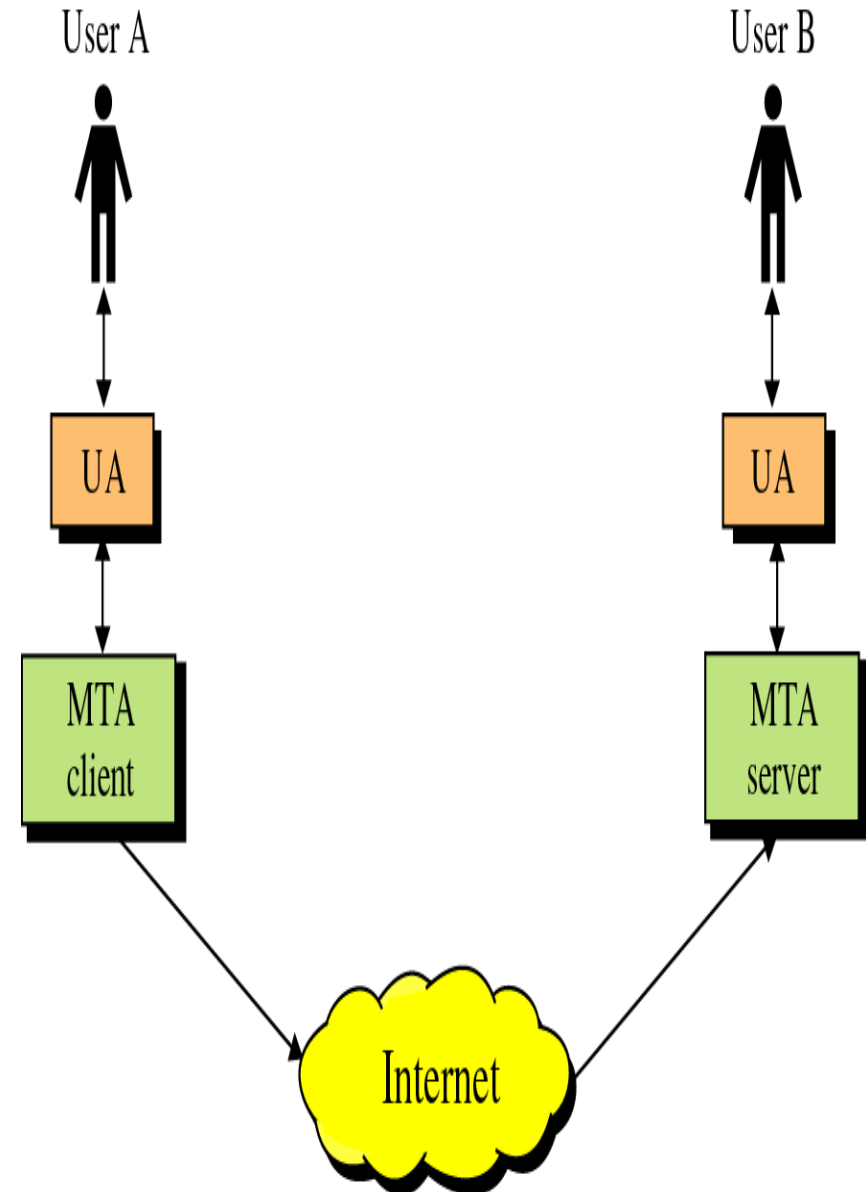
SMTP clients and servers have two main components:

User Agents (UA):

It prepares a message and encloses in an envelope.

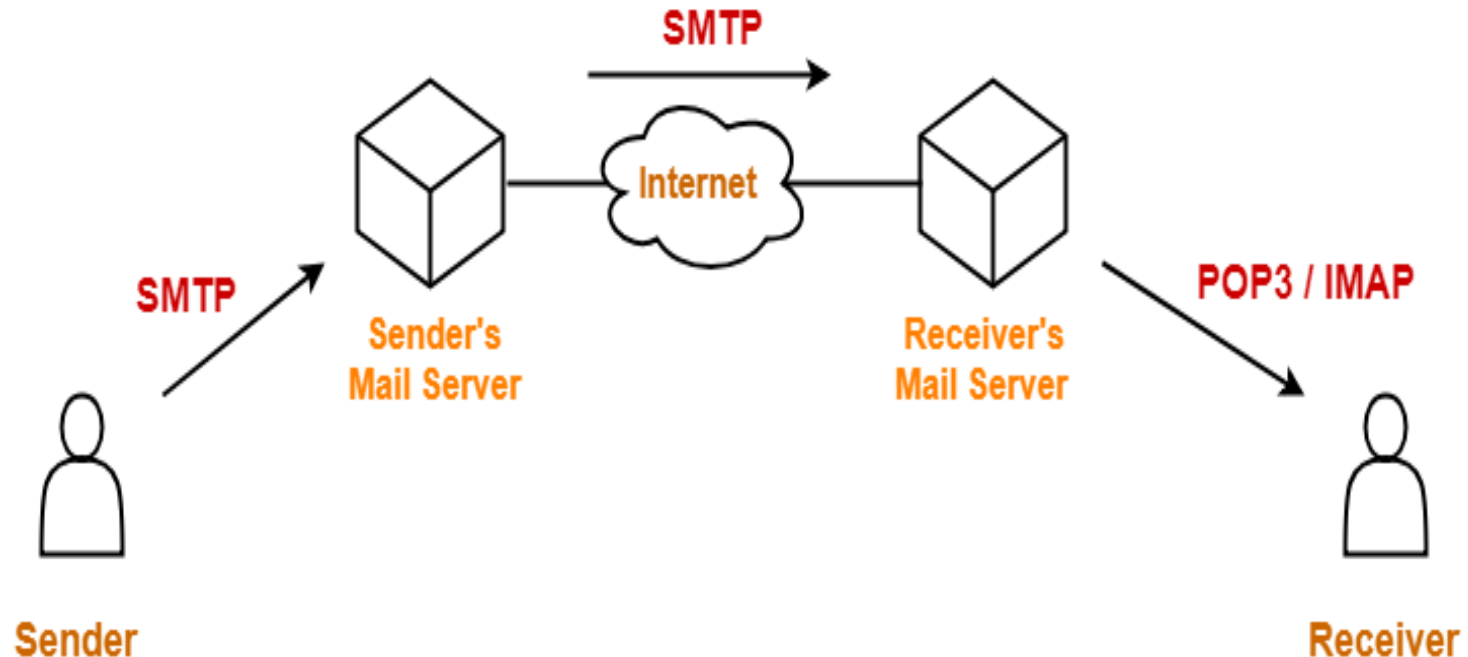
Mail Transfer Agents (MTA):

It transfers the mail across the internet.



Working SMTP-

- SMTP server is always on a listening mode.
- Client initiates a TCP connection with the SMTP server.
- SMTP server listens for a connection and initiates a connection on that port.
- The connection is established.
- Client informs the SMTP server that it would like to send a mail.
- Assuming the server is OK, client sends the mail to its mail server.
- Client's mail server use DNS to get the IP Address of receiver's mail server.
- Then, SMTP transfers the mail from sender's mail server to the receiver's mail server.



SMTP Commands

HELLO:- This command initiates the SMTP conversation.

EHELLO

This is an alternative command to initiate the conversation. ESMTP indicates that the sender server wants to use extended SMTP protocol.

MAIL FROM :- This indicates the sender's address.

RCPT TO

It identifies the recipient of the mail. In order to deliver similar message to multiple users this command can be repeated multiple times.

SIZE :- This command let the server know the size of attached message in bytes.

DATA

The **DATA** command signifies that a stream of data will follow. Here stream of data refers to the body of the message.

QUIT:- This commands is used to terminate the SMTP connection.

VERFY:- This command is used by the receiving server in order to verify whether the given username is valid or not.

EXPN:- it is same as VRFY, except it will list all the users name when it used with a distribution list.

Advantages

- Very Popular
- Supported on many platforms
- Low administration and implementation costs
- Simple addressing scheme

Limitations

- Security matters for SMTP are worse.
- Its usefulness is limited by its simplicity.
- Transmission of executable files and binary files using SMTP is not possible without converting into text files. Use MIME to send mail in other format.
- It cannot transmit text data that contains national language characters. These national language characters use 8-bit codes with values of 128 decimal or more.
- It is limited to 7-bit ASCII characters only.
- SMTP servers may reject mail messages beyond some specific length.

Post Office Protocol(POP)

POP Protocol

The POP protocol stands for Post Office Protocol. As we know that SMTP is used as a message transfer agent. When the message is sent, then SMTP is used to deliver the message from the client to the server and then to the recipient server.

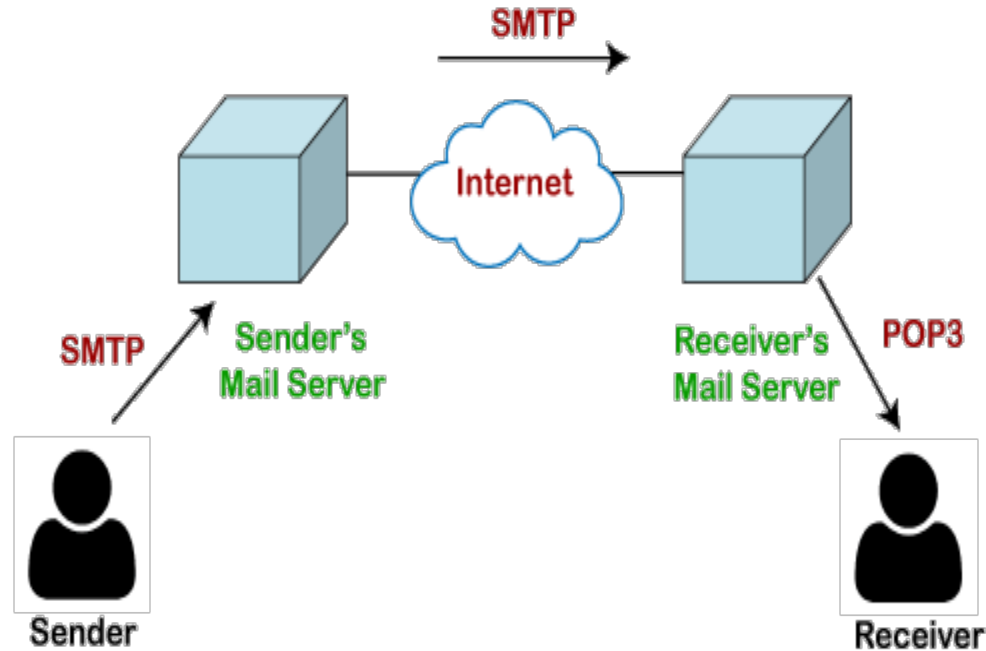
But the message is sent from the recipient server to the actual server with the help of the Message Access Agent.

The Message Access Agent contains two types of protocols, i.e., POP3 and IMAP.

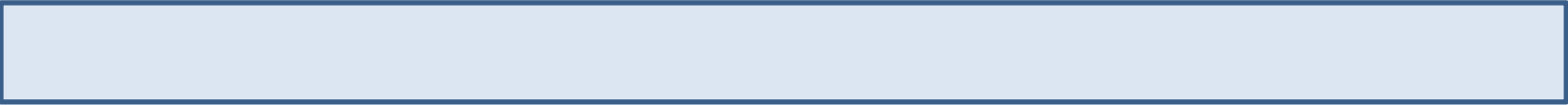
Characteristics of POP-

- POP is a pull protocol.
- POP uses TCP at the transport layer.
- POP uses port number 110.
- POP uses persistent TCP connections.
- POP is a connection oriented protocol.
- POP is an in-band protocol.
- POP is a stateful protocol until the mail is downloaded as well as stateless across sessions.

How is mail transmitted?



- Suppose sender wants to send the mail to receiver. First mail is transmitted to the sender's mail server.
- Then, the mail is transmitted from the sender's mail server to the receiver's mail server over the internet.
- On receiving the mail at the receiver's mail server, the mail is then sent to the user. The whole process is done with the help of Email protocols.



➤ The transmission of mail from the sender to the sender's mail server and then to the receiver's mail server is done with the help of the SMTP protocol.

➤ At the receiver's mail server, the POP or IMAP protocol takes the data and transmits to the actual user.

❑ Since SMTP is a push protocol so it pushes the message from the client to the server.

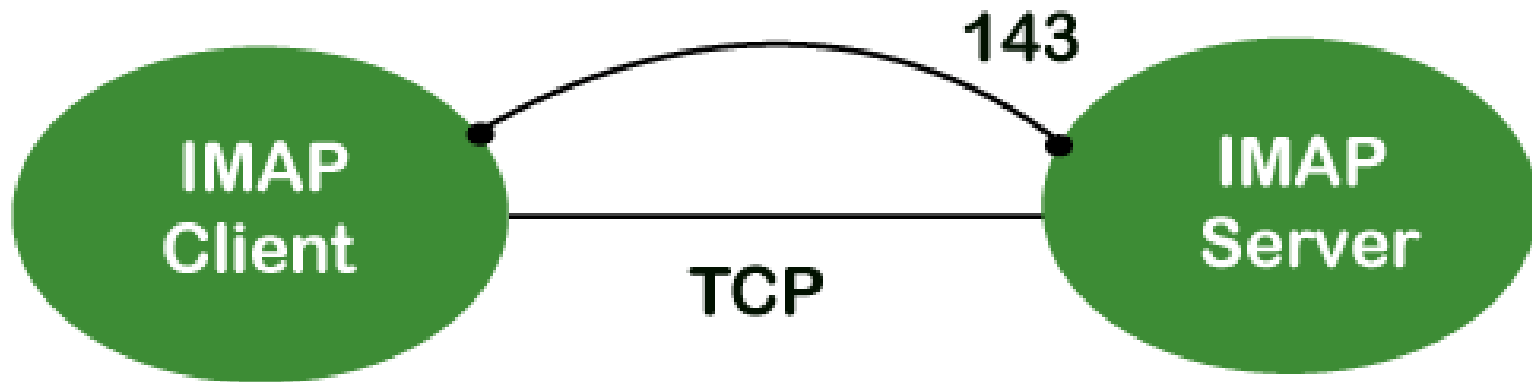
❑ As we can observe in the above figure that SMTP pushes the message from the client to the recipient's mail server.

❑ The third stage of email communication requires a pull protocol, and POP is a pull protocol.

❑ When the mail is transmitted from the recipient mail server to the client which means that the client is pulling the mail from the server.

IMAP (Internet Message Access Protocol.)

- It is an application layer protocol which is used to receive the emails from the mail server.
- It is the most commonly used protocols like POP for retrieving the emails.
- It also follows the client/server model. On one side, we have an IMAP client, which is a process running on a computer.
- On the other side, we have an IMAP server, which is also a process running on another computer.
- Both computers are connected through a network.



- The IMAP protocol resides on the TCP/IP transport layer which means that it implicitly uses the reliability of the protocol.
- Once the TCP connection is established between the IMAP client and IMAP server, the IMAP server listens to the port 143 by default, but this port number can also be changed.

By default, there are two ports used by IMAP:

Port 143: It is a non-encrypted IMAP port.

Port 993: This port is used when IMAP client wants to connect through IMAP securely.

Characteristics of IMAP-

- IMAP is a pull protocol.
- IMAP uses TCP at the transport layer.
- IMAP uses port number 143.
- IMAP uses persistent TCP connections.
- IMAP is a connection oriented protocol.
- IMAP is an in-band protocol.
- IMAP is a stateful protocol.
- IMAP distributes mail boxes across multiple servers.

Multipurpose Internet Mail Extension (MIME)

- ❖ MIME is a standard which was proposed by Bell Communications in 1991 in order to expand limited capabilities of email.
- ❖ MIME is a kind of *add on or a supplementary protocol* which allows non-ASCII data to be sent through SMTP.
- ❖ It allows the users to exchange different kinds of data files on the Internet: audio, video, images, application programs as well.

Why do we need MIME?

Limitations of Simple Mail Transfer Protocol (SMTP):

- SMTP has a very simple structure
- It's simplicity however comes with a price as it only send messages in NVT 7-bit ASCII format.
- It cannot be used for languages that do not support 7-bit ASCII format such as- French, German, Russian, Chinese and Japanese, etc. so it cannot be transmitted using SMTP. So, in order *to make SMTP more broad we use MIME*.
- It cannot be used to send binary files or video or audio data.

Purpose and Functionality of MIME –

- Growing demand for Email Message as people also want to express in terms of Multimedia. So, MIME another email application is introduced as it is not restricted to textual data.
- MIME *transforms non-ASCII data* at sender side to NVT 7-bit data and delivers it to the client SMTP.
- The message at receiver side is transferred back to the original data.
- As well as we can send video and audio data using MIME as it transfers them also in 7-bit ASCII data.

Features of MIME –

- It is able to send multiple attachments with a single message.
- Unlimited message length.
- Binary attachments (executables, images, audio, or video files) which may be divided if needed.
- MIME provided support for varying content types and multi-part messages.

Working of MIME –

- Suppose a user wants to send an email through user agent and it is in a non-ASCII format so there is a MIME protocol which converts it into 7-bit NVT ASCII format.
- Message is transferred through e-mail system to the other side in 7-bit format now MIME protocol again converts it back into non-ASCII code and now the user agent of receiver side reads it and then information is finally read by the receiver.
- MIME header is basically inserted at the beginning of any e-mail transfer.

MIME Header:

It is added to the original e-mail header section to define transformation. There are *five headers* which we add to the original header:

❖ **MIME Version** – Defines version of MIME protocol. It must have the parameter *Value 1.0*, which indicates that message is formatted using MIME.

❖ **Content Type** – Type of data used in the body of message. They are of different types like text data (plain, HTML), audio content or video content.

❖ **Content Type Encoding** – It defines the method used for encoding the message. Like 7-bit encoding, 8-bit encoding, etc.

❖ **Content Id** – It is used for uniquely identifying the message.

❖ **Content description** – It defines whether the body is actually image, video or audio.

Simple Network Management Protocol (SNMP) –

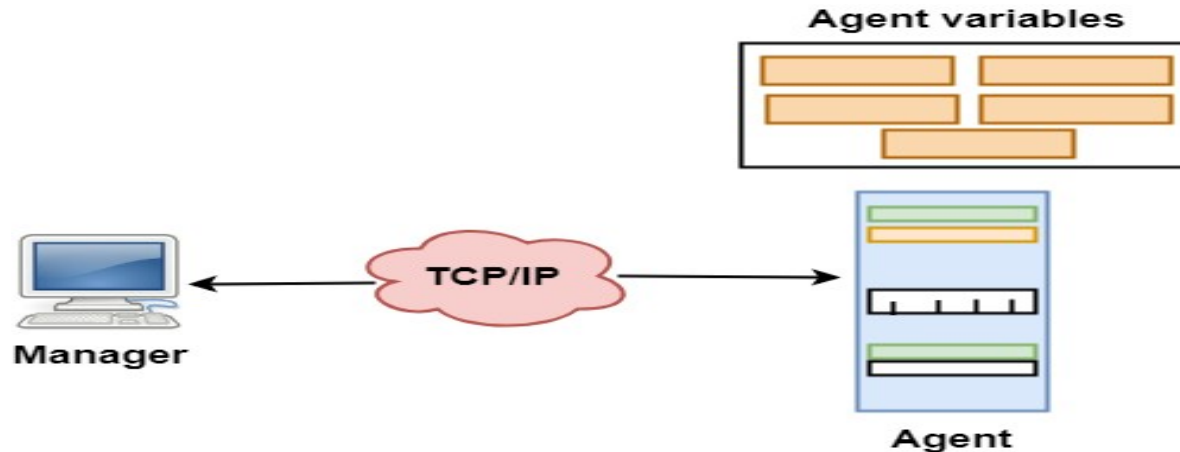
If an organization has 1000 of devices then to check all devices, one by one every day, are working properly or not is a hectic task.

To ease these up, Simple Network Management Protocol (SNMP) is used.

Simple Network Management Protocol (SNMP) –

- SNMP is an application layer protocol which uses UDP port number 161/162.
- SNMP is used to monitor the network, detect network faults and sometimes even used to configure remote devices.
- SNMP is a framework used for managing devices on the internet.
- It provides a set of operations for monitoring and managing the internet.

SNMP Concept



SNMP has two components Manager and agent.

- The manager is a host that controls and monitors a set of agents such as routers.
- It is an application layer protocol in which a few manager stations can handle a set of agents.
- The protocol designed at the application level can monitor the devices made by different manufacturers and installed on different physical networks.
- It is used in a heterogeneous network made of different LANs and WANs connected by routers or gateways.

SNMP components –

There are 3 components of SNMP:

SNMP Manager –

It is a centralized system used to monitor network. It is also known as Network Management Station (NMS)

SNMP agent –

It is a software management software module installed on a managed device. Managed devices can be network devices like PC, router, switches, servers etc.

Management Information Base –

MIB consists of information of resources that are to be managed. These information is organized hierarchically. It consists of objects instances which are essentially variables.

SNMP messages –

Different variables are:

GetRequest –

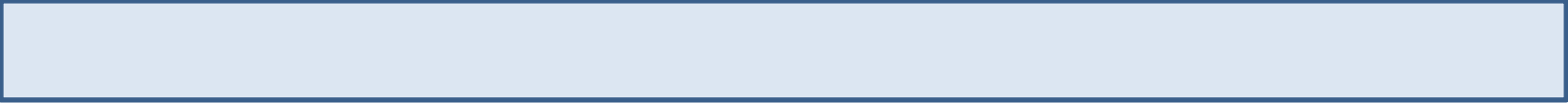
SNMP manager sends this message to request data from SNMP agent. It is simply used to retrieve data from SNMP agent. In response to this, SNMP agent responds with requested value through response message.

GetNextRequest –

This message can be sent to discover what data is available on a SNMP agent. The SNMP manager can request for data continuously until no more data is left. In this way, SNMP manager can take knowledge of all the available data on SNMP agent.

GetBulkRequest –

This message is used to retrieve large data at once by the SNMP manager from SNMP agent. It is introduced in SNMPv2c.



SetRequest –

It is used by SNMP manager to set the value of an object instance on the SNMP agent.

Response –

It is a message send from agent upon a request from manager. When sent in response to Get messages, it will contain the data requested. When sent in response to Set message, it will contain the newly set value as confirmation that the value has been set.

Trap –

These are the message send by the agent without being requested by the manager. It is sent when a fault has occurred.

InformRequest –

It was introduced in SNMPv2c, used to identify if the trap message has been received by the manager or not. The agents can be configured to set trap continuously until it receives an Inform message. It is same as trap but adds an acknowledgement that trap doesn't provide.

SNMP security levels –

It defines the type of security algorithm performed on SNMP packets. These are used in only SNMPv3. There are 3 security levels namely:

noAuthNoPriv –

This (no authentication, no privacy) security level uses community string for authentication and no encryption for privacy.

authNopriv – This security level (authentication, no privacy) uses HMAC with Md5 for authentication and no encryption is used for privacy.

authPriv – This security level (authentication, privacy) uses HMAC with Md5 or SHA for authentication and encryption uses DES-56 algorithm.